Practical – 4

AIM: There is a new barn with N stalls and C cows. The stalls are located on a straight line at positions x1,xN(0 <= xi <= 1,000,000,000). We want to assign the cows to the stalls, such that the minimum distance between any two of them is as large as possible. What is the largest minimum distance?

Source code:-

```
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    return (*(int *)a - *(int *)b);
}

int can_place_cows(int *stalls, int n, int cows, int dist) {
    int count = 1;
    int prev = stalls[0];
    for (int i = 1; i < n; i++) {
        if (stalls[i] - prev >= dist) {
            count++;
            prev = stalls[i];
        }
    }
    return count >= cows;
}
int max_min_distance(int *stalls, int n, int cows) {
    int low = 1;
    int high = stalls[n - 1] - stalls[0];

    int ans = -1;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (can_place_cows(stalls, n, cows, mid)) {
            ans = mid;
            low = mid + 1;
        } else {
            high = mid - 1;

        }
```

```
 }
    return ans;
}


int main() {
    int n, cows;
    printf("Enter the number of stalls: ");
    scanf("%d", &n);
    int *stalls = (int *)malloc(n * sizeof(int));
    printf("Enter the stalls: ");
    for (int i = 0; i < n; i++) {
        scanf("%d", &stalls[i]);
    }
    printf("Enter the number of cows: ");
    scanf("%d", &cows);
    qsort(stalls, n, sizeof(int), compare);
    printf("The maximum minimum distance is: %d\n", max_min_distance(stalls, n, cows));
    free(stalls);
    return 0;
}
```

Output:

```
Output

/tmp/NyNfA5xV97.o
Enter the number of stalls: 6
Enter the stalls: 3 6 8 9 2 5
Enter the number of cows: 3
The maximum minimum distance is: 3


=== Code Execution Successful ===
```